Assignment 2 | End-to-end classification and deployment

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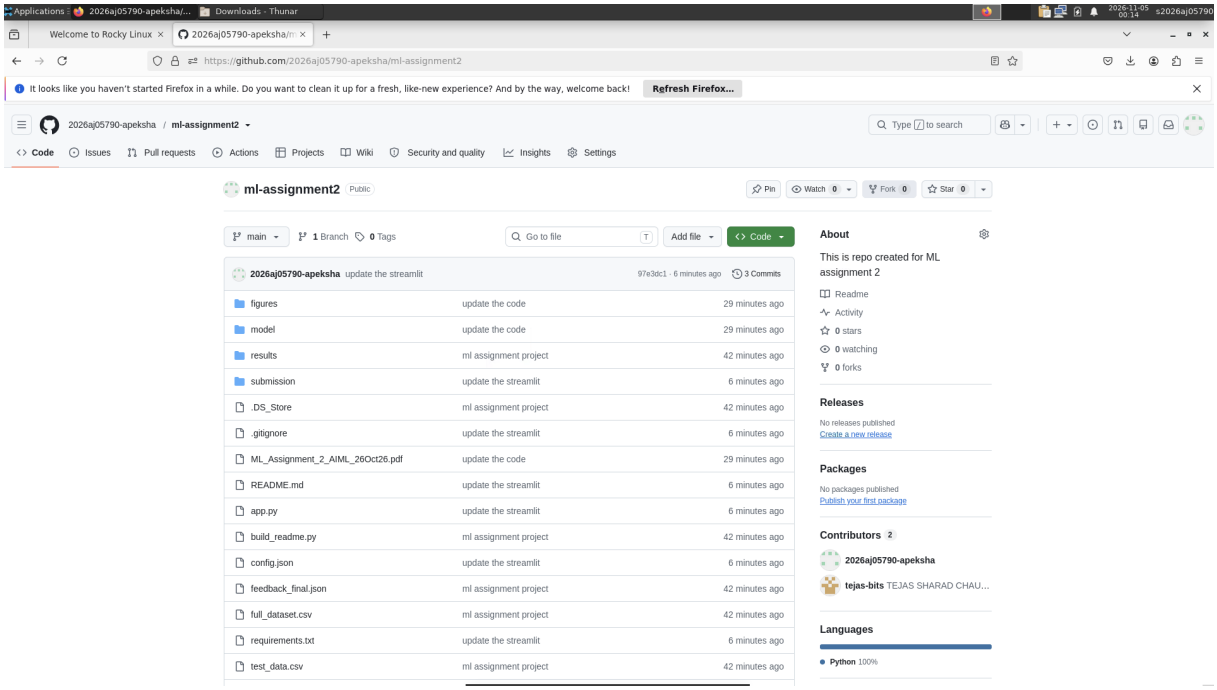
1. GitHub Repository Link

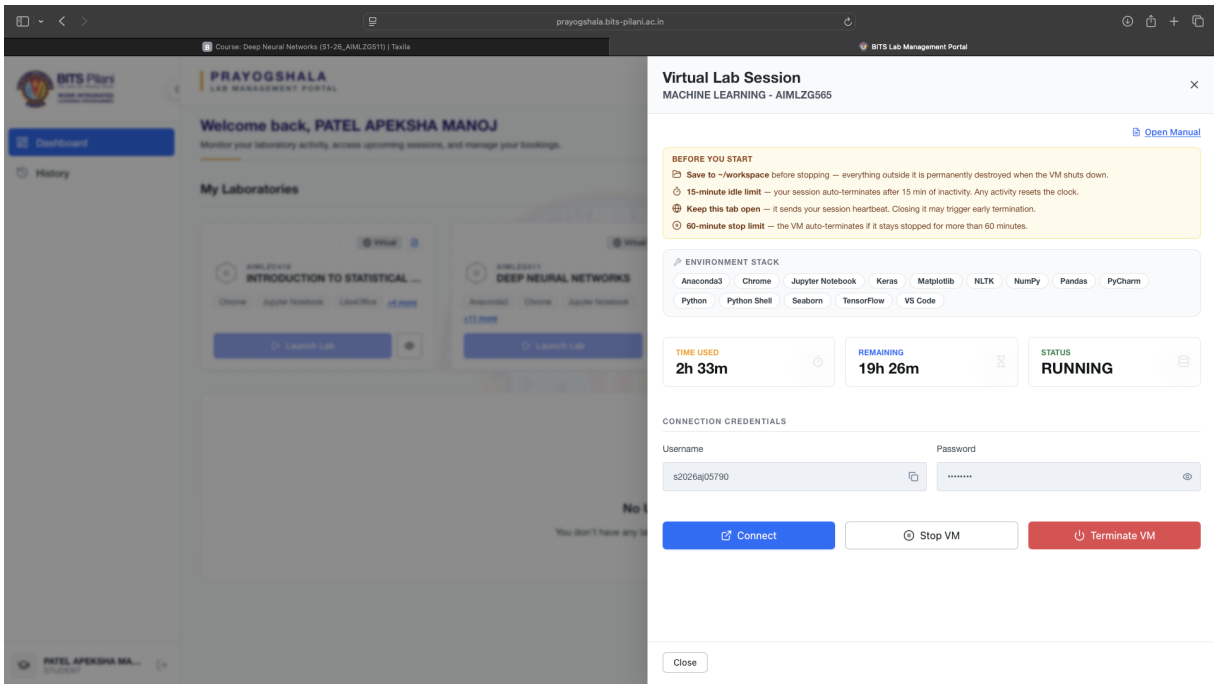
https://github.com/2026aj05790-apeksha/ml-assignment2

2. Live Streamlit App Link

https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

3. BITS Virtual Lab usage screenshots




prayogshala.bits-pilani.ac.in
Course: Deep Neural Networks (S1-26_AIMLZG511) | Taxila
BITS Lab Management Portal
PRAYOGSHALA
LAB MANAGEMENT PORTAL
Dashboard
History
Welcome back, PATEL APEKSHA MANOJ
Monitor your laboratory activity, access upcoming sessions, and manage your bookings.
My Laboratories
INTRODUCTION TO STATISTICAL ...
DEEP NEURAL NETWORKS
Launch Lab
Virtual Lab Session
MACHINE LEARNING - AIMLZG565
Open Manual
BEFORE YOU START
Save to ~/workspace before stopping — everything outside it is permanently destroyed when the VM shuts down.
15-minute idle limit — your session auto-terminates after 15 min of inactivity. Any activity resets the clock.
Keep this tab open — it sends your session heartbeat. Closing it may trigger early termination.
60-minute stop limit — the VM auto-terminates if it stays stopped for more than 60 minutes.
ENVIRONMENT STACK
Anaconda3
Chrome
Jupyter Notebook
Keras
Matplotlib
NLTK
NumPy
Pandas
PyCharm
Python
Python Shell
Seaborn
TensorFlow
VS Code
TIME USED
2h 33m
REMAINING
19h 26m
STATUS
RUNNING
CONNECTION CREDENTIALS
Username
Password
s2026aj05790
Connect
Stop VM
Terminate VM
Close

4. README Content

■ Breast-Cancer Diagnosis – Five-Classifier Comparison & Streamlit Deployment

BITS Pilani WILP · M.Tech (AIML) · Machine Learning · Assignment 2

a. Problem statement

Early, accurate discrimination between **malignant** and **benign** breast tumours from fine-needle-aspirate (FNA) cell-nucleus measurements reduces unnecessary biopsies and prevents missed cancers.

The task is a **binary classification** problem: build, evaluate and compare five classifiers (Logistic Regression, Decision Tree, KNN, Naive Bayes, SVM) on one dataset using six metrics

(Accuracy, AUC, Precision, Recall, F1, MCC), with malignant (class 0) consistently treated as the positive clinical condition. The interactive Streamlit app evaluates the *saved* models on an uploaded test CSV without retraining.

b. Dataset description

Item	Detail
Source	Breast Cancer Wisconsin (Diagnostic) – UCI Machine Learning Repository (Wolberg, Street & Mangasarian, 1995), loaded via `sklearn.datasets.load_breast_cancer`
Instances	569 (train 455 / test 114, stratified 80:20, seed 42)
Features	30 numeric (>= 12 required): 10 nucleus measurements × {mean, standard error, worst}
Target	`diagnosis` – 0 = malignant (212, 37.3 %), 1 = benign (357, 62.7 %)
Class balance	Moderately imbalanced (1 : 1.68) → MCC and recall reported alongside accuracy
Missing values	None
Pre-processing	`StandardScaler` inside each sklearn `Pipeline` (fit on training data only – no leakage); the saved `.joblib` therefore contains the complete preprocessing

The 10 base measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension.

Many are strongly collinear (radius–perimeter–area r > 0.95), which matters for Naive Bayes (see observations).

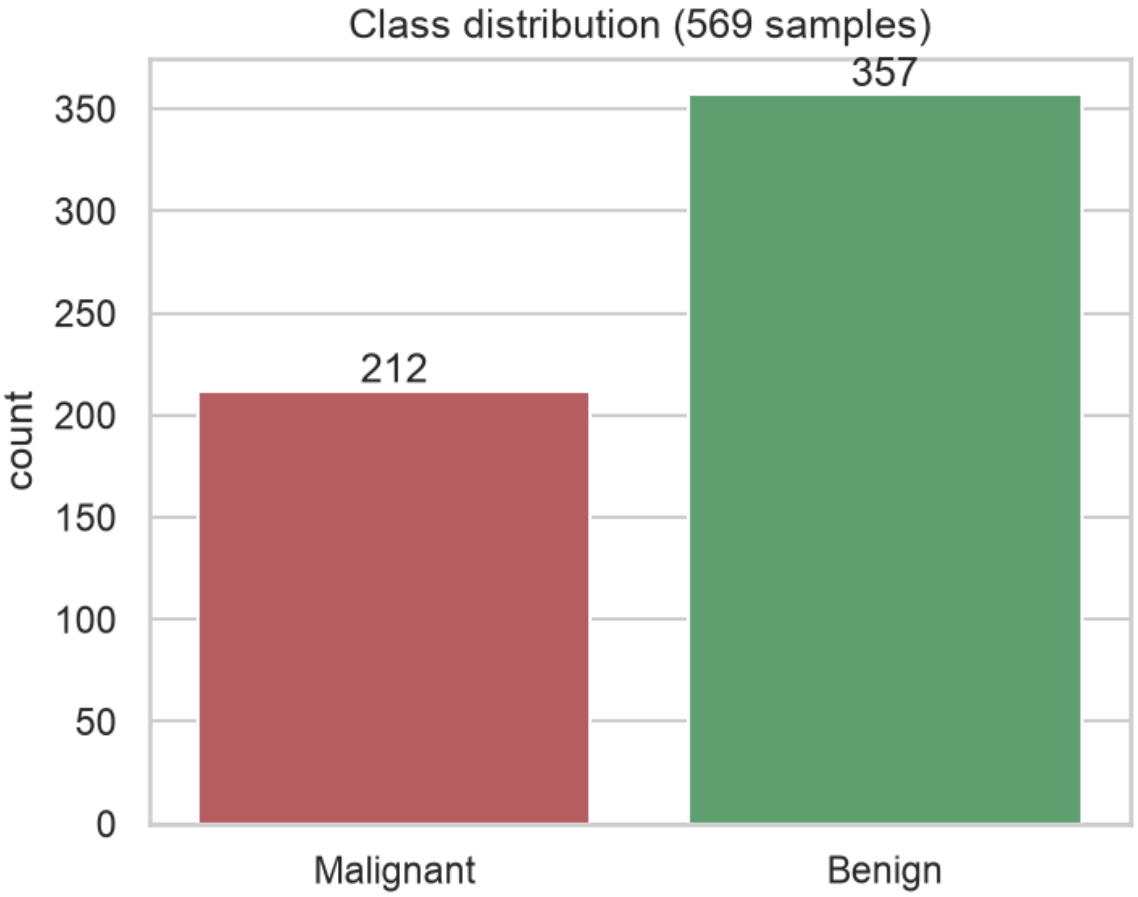


Figure 1 – Class distribution

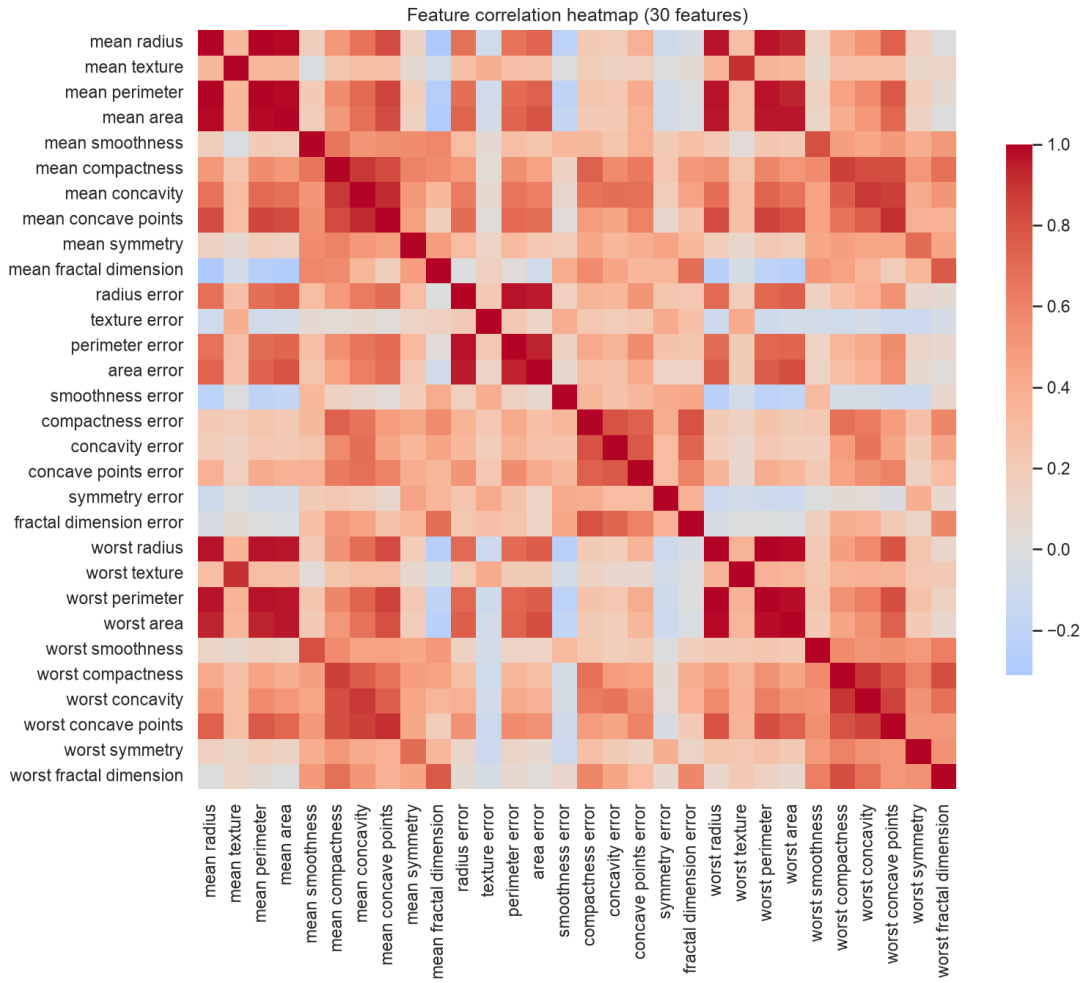


Figure 2 – Feature correlation heatmap: blocks of highly correlated size features

c. GitHub Repository Link

https://github.com/2026aj05790-apeksha/ml-assignment2

```
project-folder/
|-- app.py                # root Streamlit launcher used for deployment
|-- train.py              # root training launcher used by the assignment
|-- build_readme.py       # root documentation launcher
|-- full_dataset.csv      # rubric-required full-data mirror
|-- src/                  # application, training, and README-generation code
|-- artifacts/            # generated data, models, metrics, and figures
|-- test_data.csv         # rubric-required hold-out test data mirror
|-- model/                # rubric-required saved-model mirror
|-- problem_statement/    # original assignment PDF
|-- submission/           # final PDF and BITS Lab screenshots
|-- config/               # student, link, and submission metadata
|-- reports/              # grader feedback and reports
`-- requirements.txt
```

The root `app.py`, `train.py`, and `build_readme.py` files are thin compatibility launchers; implementation remains in `src/`. `artifacts/` is the generated source-of-truth output, while root `model/`, `test_data.csv`, and `full_dataset.csv` are refreshed compatibility copies required by the assignment layout.

Live Streamlit app: https://ml-assignment2-bxkybtx2fackxbtb4bh9vk.streamlit.app

How to run / reproduce

```
pip install -r requirements.txt
python train.py           # retrains all models and regenerates artifacts/
streamlit run app.py      # launches the evaluation app
```

d. Models used

All five models are trained on the **same** 455-row training split and evaluated on the **same** 114-row hold-out test set.

Hyper-parameters: LR (C=1), Decision Tree (max_depth=5, min_samples_leaf=3), KNN (k=7, distance-weighted), Gaussian NB (defaults), SVM (RBF, C=1, gamma='scale').

Comparison table (test set)

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9825	0.9954	0.9762	0.9762	0.9762	0.9623
Decision Tree	0.9035	0.9373	0.8444	0.9048	0.8736	0.7969
KNN	0.9737	0.9884	1.0000	0.9286	0.9630	0.9442
Naive Bayes	0.9298	0.9868	0.9048	0.9048	0.9048	0.8492
Support Vector Machine	0.9825	0.9950	0.9762	0.9762	0.9762	0.9623

Generalisation check (5-fold stratified CV on training data vs train vs test)

ML Model Name	Train Acc	5-fold CV Acc	Test Acc
Logistic Regression	0.9890	0.978 ± 0.010	0.9825
Decision Tree	0.9802	0.916 ± 0.019	0.9035
KNN	1.0000	0.963 ± 0.020	0.9737
Naive Bayes	0.9385	0.934 ± 0.029	0.9298
Support Vector Machine	0.9824	0.969 ± 0.015	0.9825

Additional diagnostic metrics

The required six metrics are shown above. For clinical interpretation, malignant (class 0) is treated as the positive condition in this section. Malignant sensitivity measures cancers correctly identified, and malignant FNR measures cancers predicted benign. Benign specificity measures the benign cases correctly classified as benign.

ML Model Name	Balanced Accuracy	Malignant Sensitivity	Benign Specificity	Malignant FNR
Logistic Regression	0.9812	0.9861	0.9762	0.0139
Decision Tree	0.9038	0.9028	0.9048	0.0972
KNN	0.9643	1.0000	0.9286	0.0000
Naive Bayes	0.9246	0.9444	0.9048	0.0556
Support Vector Machine	0.9812	0.9861	0.9762	0.0139

Clinically labelled confusion counts

ML Model Name	Malignant → malignant	Malignant → benign (missed cancer)	Benign → malignant (false alarm)	Benign → benign
Logistic Regression	71	1	1	41
Decision Tree	65	7	4	38
KNN	72	0	3	39
Naive Bayes	68	4	4	38
Support Vector Machine	71	1	1	41

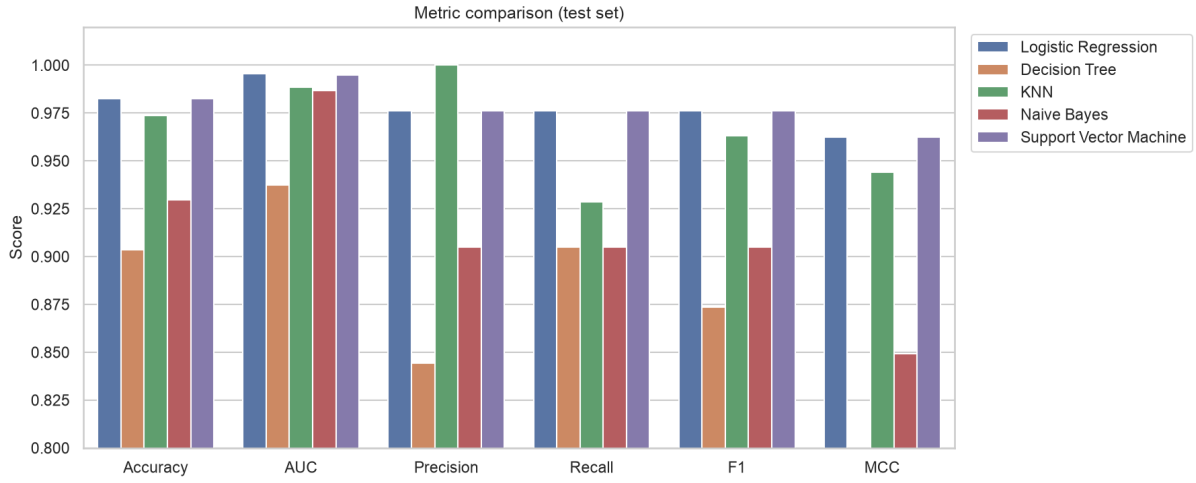


Figure 3 – Metric comparison across models

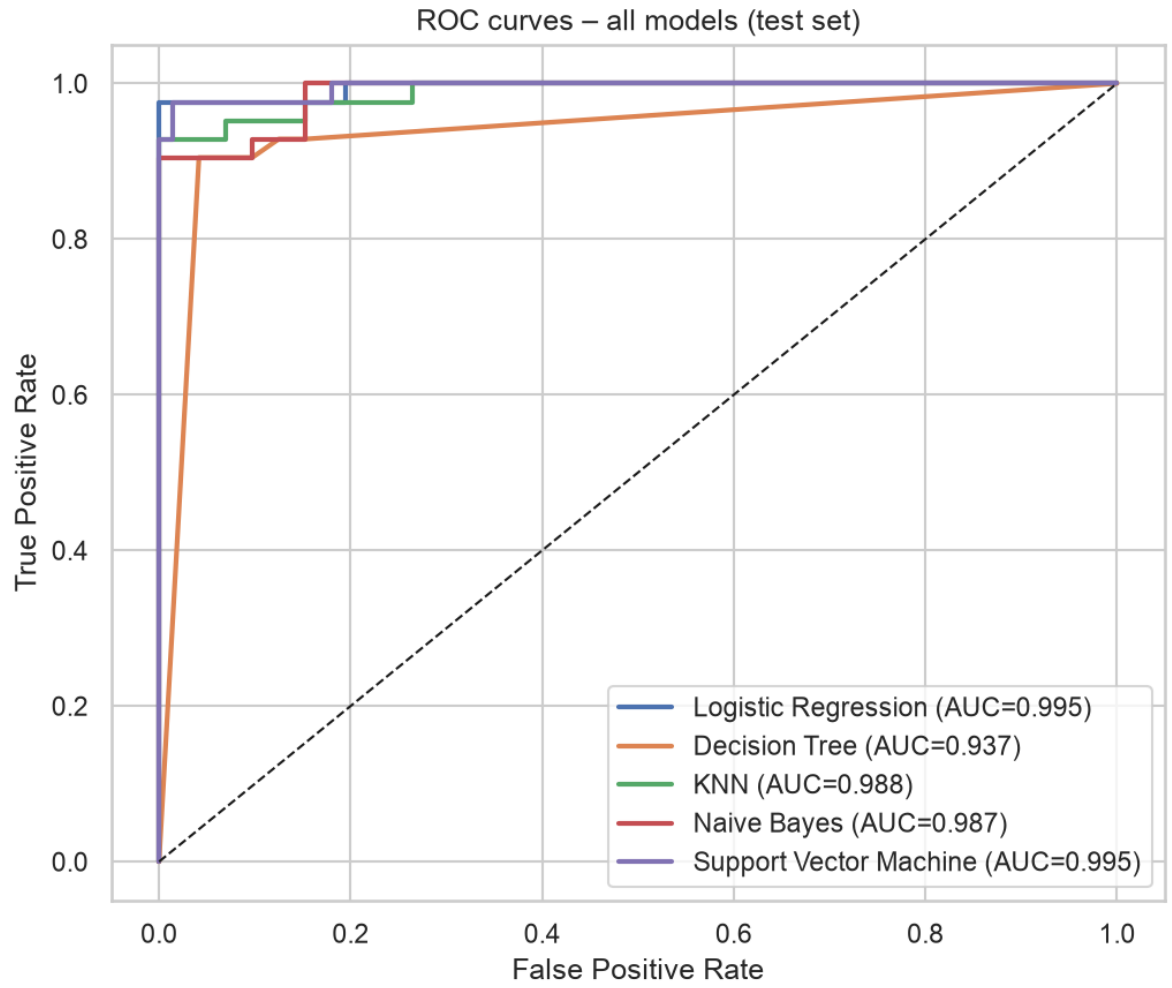


Figure 4 – ROC curves (test set)

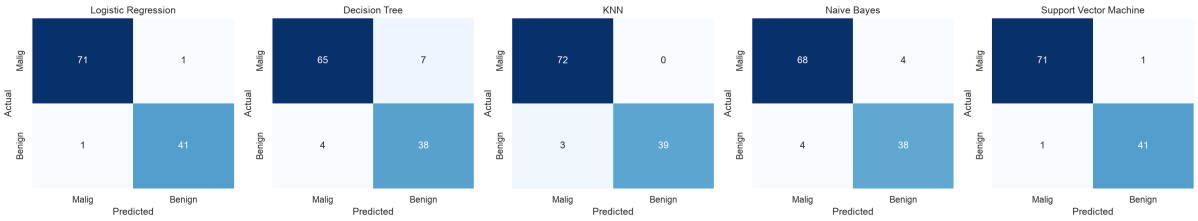


Figure 5 – Confusion matrices (class 0 = malignant)

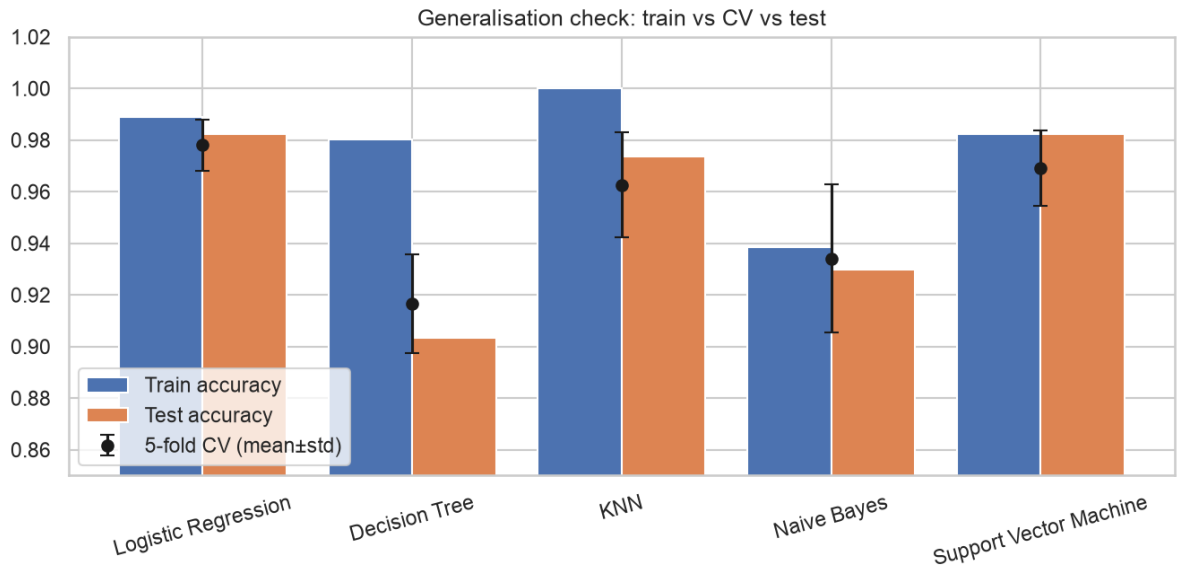


Figure 6 – Train vs CV vs test accuracy

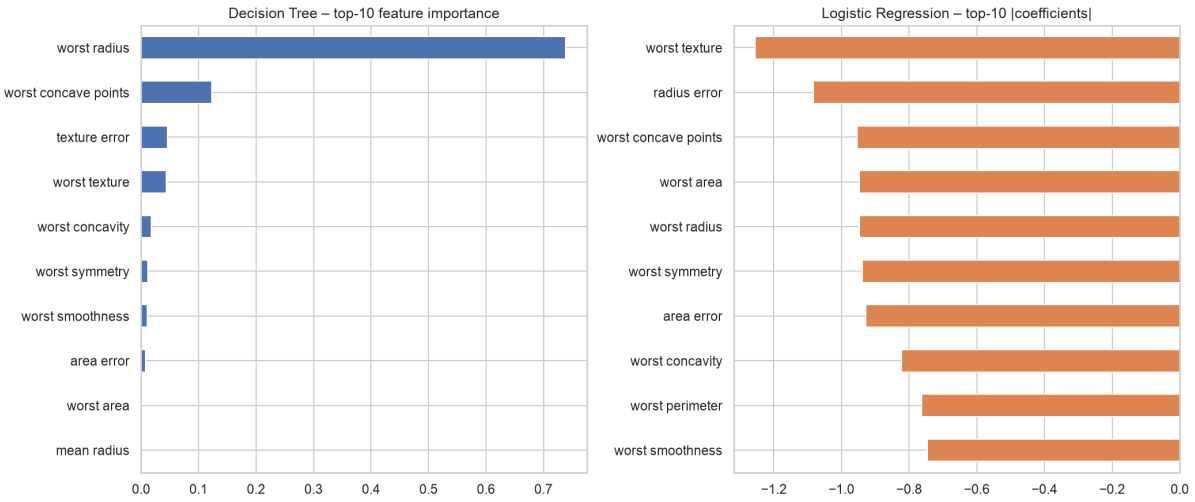


Figure 7 – Feature importance (Decision Tree) and coefficients (Logistic Regression)

Observations on model performance

ML Model Name	Observation about model performance
Logistic Regression	Logistic Regression records 98.25% accuracy, AUC 0.9954, and MCC 0.962. It makes 1 missed-cancer error and 1 benign false alarm on the 114-row test set. Its train score (0.989) is close to the cross-validation result (0.978 ± 0.010), while the linear coefficients provide useful directionality for interpretation.
Decision Tree	The Decision Tree has the lowest test performance: 90.35% accuracy, AUC 0.937, and MCC 0.797. Its training accuracy (0.980) is much higher than its test accuracy, indicating variance and overfitting. It misses 7 malignant cases, the largest missed-cancer count, although its rule structure remains easy to explain.
KNN	KNN reaches 97.37% accuracy and the best malignant sensitivity (1.000) because it misses no malignant test cases. The trade-off is 3 benign cases flagged as malignant, which lowers benign specificity to 0.929. Standardisation is essential for distance calculations; its training score of 1.0000 is higher than its 5-fold CV score (0.963 ± 0.020), showing that memorisation does not transfer perfectly.
Naive Bayes	Naive Bayes obtains 92.98% accuracy and MCC 0.849, while its AUC is relatively strong at 0.9868. The gap suggests useful ranking but less reliable threshold decisions. Highly correlated radius, perimeter, and area variables conflict with the independence assumption, and the similar train (0.939) and test scores point more to bias than overfitting.
Support Vector Machine	The RBF SVM is effectively tied with Logistic Regression: accuracy 98.25%, AUC 0.9950, and MCC 0.962. Its CV accuracy (0.969 ± 0.015) is below Logistic Regression's, so the additional non-linear flexibility does not improve this split despite the strong test result.
Overall, Winner for your dataset?	**Logistic Regression** (with SVM as a statistical tie). LR and SVM have identical test accuracy (98.25%), MCC (0.962) and recall, but LR has the higher 5-fold CV accuracy (0.978 ± 0.010 vs 0.969 ± 0.015), the highest AUC, trains in milliseconds and is interpretable. We use MCC and malignant sensitivity alongside accuracy because the malignant class is the safety-critical class; a malignant tumour predicted benign is the clinically costly error. KNN avoids missed malignant cases here but raises more benign false alarms. With only 114 test rows, one sample changes accuracy by 0.88%, so the LR/SVM tie should not be over-interpreted.

Error analysis

15 of 114 test samples are misclassified by at least one model and **1** by all five – these are genuinely borderline tumours whose measurements overlap the other class.

Most of the accuracy gap between models therefore comes from a small set of ambiguous cases, which is why MCC/recall and CV are used in addition to a single test split.

Streamlit application features

CSV upload (test data only) · model dropdown · six required metric cards · detailed per-class classification report · confusion matrix · specificity · balanced accuracy · ROC curve · adjustable decision threshold ·

side-by-side comparison of all five saved models on the uploaded data · prediction download · input validation (missing columns / non-binary target).

Models are loaded with `joblib`; **no training happens in the app**.

Limitations

• Single public dataset with 569 rows; a 114-row test set gives wide confidence intervals (one sample ≈ 0.88 % accuracy), so LR vs SVM differences are not statistically significant.

• Hyper-parameters were set by reasoned defaults and light CV, not an exhaustive search; data come from one institution, so external generalisation is unverified.

• Models are decision-support tools, not diagnostic devices.

Future work

• Nested cross-validation with grid/Bayesian search, probability calibration (Platt/isotonic) – especially for Naive Bayes.

• Ensembles (Random Forest, Gradient Boosting, stacking), SHAP explanations, external validation on another cohort.

• Cost-sensitive threshold selection to explicitly minimise false negatives.